

SYSTEM ARCHITECTURE & ENGINEERING SPECIFICATION

DOCURE Technology Stack

An in-depth technical analysis of the Artificial Intelligence models, frontend subsystems, cloud databases, mapping services, and background daemon engineering powering DOCURE.

PROJECT: DOCURE-AI VERSION: 2.4.0-PROD AUTHOR: Debug-Creww DATE: August 2026

1. Executive System Overview

DOCURE is an intelligent, privacy-conscious clinical healthcare assistant designed to bridge patient triage, AI-driven pathology report diagnostics, real-time medical facility dispatching, and automated medication schedule compliance. The application operates as a high-performance Single Page Application (SPA) leveraging edge AI models, real-time spatial APIs, and serverless persistence.

2. Architectural Layer Matrix

Frontend Layer

Modern React 18 component tree with fast Vite HMR bundling and modular node architecture.

- React 18.2
- Vite 5.4
- Tailwind CSS 3.4
- Lucide React
- Canvas EKG API

AI & Vision Engine

NVIDIA NIM inference API with Meta Llama 70B text model & 90B Vision multimodal OCR.

- NVIDIA NIM
- Llama 3.1 70B
- Llama 3.2 90B Vision
- PDF.js

Cloud Database & Backend

Serverless Firestore document storage with real-time websocket snapshots and audit logging.

- Firebase Firestore
- Firebase SDK v10
- Vite Proxy API
- Google Calendar Sync

GIS & Medical Radar

Real-time TomTom POI search, Leaflet.js interactive maps, GPS geolocation, and driving times.

- TomTom Search API
- Leaflet.js 1.9
- OpenStreetMap
- Haversine GIS



3. NVIDIA AI Lab Report Analyzer Pipeline

The Pathology OCR engine is built on a **dual-strategy architecture** that dynamically switches between client-side digital text extraction and multimodal visual transformer parsing:

```
[Step 1: Upload] User uploads Diagnostic PDF or Photo (Dr LaL / SRL / Thyrocare)
└─ IF PDF: Mozilla PDF.js parses binary text & row coordinates (lossless text)
└─ IF IMAGE: Canvas downscales to 1280px & converts to Base64 JPEG payload
[Step 2: NIM API] Proxied through /api/nvidia to prevent CORS & securely sign key
└─ PDF Route: Meta Llama 3.1 70B Instruct (Prompt: Serial Biomarkers Extraction)
└─ Image Route: Meta Llama 3.2 90B Vision Instruct (Visual layout OCR)
[Step 3: Classifier] Balanced JSON parser categorizes biomarkers into HIGH, NORMAL, LOW
[Step 4: Insights] Produces Health Score (0-100), positive insights, risks & specialist referral.
```

Pipeline Component	Technology Used	Role in System
Text LLM Model	meta/llama-3.1-70b-instruct	Processes dense lab texts, computes clinical ranges, formats serial JSON.
Vision LLM Model	meta/llama-3.2-90b-vision-instruct	Scans image tables, letterheads, and photographed medical papers.
Client PDF Parser	PDF.js (pdfjs-dist 3.11)	In-browser canvas rendering and coordinate-grouped text line reconstruction.
Reverse Proxy	Vite HTTP Proxy Middleware	Routes localhost calls to integrate.api.nvidia.com without browser CORS blocks.



4. Real-Time Medical Radar & Spatial GIS

The **Medical Radar (Node 03)** allows users to discover verified healthcare providers within a 5–15 km radius, complete with direct emergency calling and Google Maps turn-by-turn routing:



TomTom POI Geosearch

Queries TomTom Places API v2 with categories: `7321` (Hospitals), `7322` (Doctors), `7324` (Clinics), `7326` (Pharmacies). Automatic fallback to hardcoded verified Ghaziabad/NCR directory if offline.



Leaflet.js Mapping Engine

Renders custom HTML5 pulsing DivIcons with category color coding: Teal (Hospitals), Blue (Doctors), Purple (Diagnostic Labs), and Green (Pharmacies).

Facility Category	Badge Color	Emoji	Primary Action
Hospitals / Trauma Centers	Teal (<code>#0f766e</code>)		1-Click Direct Hotline + Emergency Ambulance Route
Doctors / Specialist Clinics	Royal Blue (<code>#2563eb</code>)		Specialist Consult Appointment Call
Diagnostic Labs	Violet (<code>#8b5cf6</code>)		Sample Collection & Blood Test Dispatch
24/7 Pharmacies	Emerald (<code>#10b981</code>)		Prescription Fulfillment & Emergency Drugs



5. Medication Scheduler & Background Daemon

The reminder subsystem is engineered as an autonomous background listener that ensures patients never miss scheduled dosages:



Cloud Firestore

Stores user medications under `users/{uid}/reminders`. Synchronizes instantaneously across tabs via `onSnapshot`.



Polling Daemon

A background worker wakes up every 20 seconds, evaluates current time against scheduled dose hours, and triggers alarms.



In-App Toast & Audio

Synthesizes soft multi-frequency soundwaves with Web Audio API and presents actionable reminder cards with dose logging.



6. Web Audio Synthesis & Emergency Override

DOCURE includes browser-native multi-media pipelines that require zero external audio assets:



Emergency Siren Synthesizer

Generates dual-frequency ambulance wails (850Hz & 650Hz alternate square waves) using `AudioContext.createOscillator()`. Initiates countdown to dispatch coordinates to pre-saved emergency links.



Speech Recognition & TTS

Integrates browser `webkitSpeechRecognition` for hands-free voice triage in Hindi/English, alongside `window.speechSynthesis` for reading clinical answers aloud.

 7. Complete Dependency Manifest

Package	Version	Type	Function
react / react-dom	^18.2.0	Core	Virtual DOM, reactive state hooks, lifecycle reconciliation.
vite	^5.4.21	Bundler	Lightning-fast ESBuild dev server and Rollup production compiler.
tailwindcss	^3.4.0	Styling	Utility-first responsive glassmorphic UI design framework.
firebase	^10.8.0	Database	NoSQL cloud persistence and realtime websocket subscriptions.
lucide-react	^0.344.0	Icons	Clean, medical & clinical vector icon system.
leaflet	^1.9.4	Mapping	Interactive tile rendering and hardware-accelerated markers.
pdfjs-dist	3.11.174	Document Parser	Client-side binary PDF text extraction and Canvas rasterization.